Weather-Aware Multi-Sensor Perception in Simulation: A Comparative Study of CARLA and Real-World Radar Data

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***Abstract*—Autonomous driving systems rely on cameras, LiDAR, and radar to maintain perception reliability under diverse environmental conditions. Weather conditions such as fog and rain affect these modalities differently: cameras are degraded by optical visibility loss, LiDAR can suffer from scattering and range attenuation, and radar is expected to remain comparatively robust due to longer wavelength propagation. This project evaluates whether weather conditions can be inferred from sensor data collected in simulation, and whether simulated sensor outputs contain physically meaningful weather-dependent variation. We collect synchronized camera, LiDAR, and radar data in CARLA across clear, fog, and rain regimes, construct statistical feature representations for each modality, and evaluate weather classification using simple multilayer perceptron models. Empirical results indicate that while camera-derived features achieve high classification accuracy, CARLA-simulated LiDAR and radar features consistently perform randomly. This suggests a significant disparity between visual rendering and physical sensor degradation in the default CARLA environment. To test whether this failure reflects a sensor limitation or a simulation limitation, we repeat the radar-based experiment on real-world RADIATE radar data. RADIATE radar features achieve substantially higher validation accuracy than CARLA radar under the same feature construction, suggesting that real radar contains measurable weather-dependent signal that is not captured by the default CARLA radar model. The results indicate that simulation-based weather-aware perception research must explicitly account for sensor degradation fidelity before using synthetic sensor streams for weather classification or perception reliability estimation.**

***Index Terms*—adverse weather, autonomous driving, CARLA, radar, LiDAR, sensor fusion, RADIATE, simulation fidelity**

I. Introduction

Autonomous driving systems are increasingly expected to operate outside limited fair-weather operational design domains (ODDs). Expansion into regions with fog, rain, snow, freezing rain, and wet road surfaces raises two related questions: whether the vehicle should adapt its behavior under a given condition, and whether perception modules remain reliable enough for continued operation. Consequently, a robust weather-aware autonomy stack must transcend categorical labeling to incorporate a quantitative assessment of environmental impacts on sensor reliability and perception confidence.

Modern autonomous platforms address adverse weather through overlapping sensing modalities. Cameras capture semantic texture and visual appearance, LiDAR provides high-resolution geometric structure, and radar provides longer-wavelength sensing that is often more robust to fog and rain. Industry systems also increasingly combine multi-modal sensing with sensor cleaning, redundancy, and ODD-aware decision logic. For example, Waymo describes its autonomous stack as using cameras, radar, and LiDAR as complementary modalities in inclement weather, together with cleaning systems to keep sensors operational [6], [7]. These systems motivate the central assumption of this project: weather should induce measurable changes in sensor outputs, and those changes should be usable for environmental condition estimation.

Simulation environments such as CARLA are commonly used to prototype perception and control pipelines because they provide configurable weather, traffic, maps, and sensor suites [1]. However, visual weather effects and physical sensor degradation are not necessarily equivalent. A simulator may render fog and rain in RGB images while failing to modify LiDAR or radar outputs in a physically realistic manner. This distinction is important because perception models trained on synthetic data may learn misleading modality behavior if weather affects some simulated sensors but not others.

This project began with the objective of estimating weather type and severity from synchronized CARLA camera, LiDAR, and radar data. During experimentation, the problem was reformulated into a simulation fidelity question: do CARLA LiDAR and radar outputs contain weather-dependent signal at all? To answer this, we perform controlled experiments on CARLA sensor data, conduct feature ablations for radar, compare camera-only and fusion performance, and validate the simulation-based findings against real-world radar data from RADIATE [2]. The final contribution is a reproducible analysis showing that CARLA radar and LiDAR features are not discriminative for weather in the tested configuration, while real radar data exhibits measurable separability.

II. Background and Motivation

A. Weather Effects on Driving Sensors

Weather affects sensing through modality-specific physical mechanisms. Camera systems are affected by visibility reduction, reduced contrast, specular reflections, blur, and occlusion from droplets or snow. The ACDC dataset provides a benchmark for semantic perception under adverse visual conditions including fog, rain, nighttime, and snow [3]. It highlights that perception performance varies significantly across environmental conditions due to changes in visibility,

contrast, and illumination. Prior work on adverse-condition datasets such as ACDC demonstrates that models trained on standard conditions often degrade when exposed to adverse domains [3].

LiDAR degradation under rain, fog, and snow has been studied through attenuation and backscatter models. Goodin et al. developed a mathematical model for the influence of rain rate on LiDAR performance and integrated it into ADAS simulation [5]. Other multimodal studies show that fog and snow can reduce object detection performance and alter sensor reliability, motivating explicit modeling of sensor degradation rather than assuming static sensor behavior [4]. These mechanisms suggest that LiDAR point density, range distribution, and return intensity are expected to vary under physically realistic adverse weather.

Radar is expected to be more robust under adverse weather because millimeter-wave sensing is less sensitive to optical visibility loss. RADIATE was introduced as a radar dataset for automotive perception in bad weather, including multiple adverse driving conditions [2]. Radar is therefore an important modality for evaluating whether real-world weather-dependent signal is present even when simulated radar fails.

B. Weather Awareness in Deployed Systems

Weather awareness in deployed autonomous systems is generally addressed using multiple complementary strategies: direct sensor redundancy, sensor cleaning, weather-aware perception training, and ODD constraints. Commercial systems describe the use of radar, LiDAR, and cameras with overlapping fields of view to support operation in unexpected or inclement weather [6], [7]. These systems do not rely on a single sensor stream for weather robustness; instead, they combine sensing diversity with operational restrictions and infrastructure-level condition monitoring.

From a research perspective, this motivates a distinction between *weather classification* and *sensor reliability estimation*. A weather classifier predicts a label such as clear, fog, or rain. A reliability estimator attempts to determine whether sensor outputs are degraded enough to affect perception confidence or ODD validity. The current project is positioned between these two goals. The experiments first test whether weather can be classified from sensor-level features; the resulting failures and successes then indicate which modalities contain usable reliability signals.

C. Research Gap

Existing work has addressed adverse weather perception datasets, LiDAR degradation models, multimodal fusion, and real-world radar datasets. However, there remains a practical gap when simulation is used for weather-aware perception: configurable simulator weather does not guarantee physically meaningful degradation in all sensor modalities. This project evaluates that gap directly by comparing CARLA camera, LiDAR, and radar outputs against real-world RADIATE radar data.

III. Design

A. Simulation Environment

CARLA was used as the primary simulation environment. Experiments were conducted in Town03 using synchronous mode with a fixed simulation step of 0.1 seconds. Weather was controlled using CARLA weather parameters including cloudiness, precipitation, precipitation deposits, wind intensity, fog density, fog distance, fog falloff, wetness, and sun altitude angle. The initial data collection configuration used two ego vehicles operating under autopilot, with synchronized sensor capture per simulation tick.

Each ego vehicle was equipped with a roof-mounted ray-cast LiDAR, radar sensors, and a front-facing RGB camera. The initial LiDAR configuration used 64 channels, 100 m range, high points-per-second density, and a 10 Hz rotation frequency. Radar data was collected as sparse detections containing depth, azimuth, altitude, and velocity. Camera data was stored as RGB images. All modalities were indexed by frame identifier to support synchronized feature extraction and later qualitative inspection.

B. Evolution of the Sensor Setup

The project began with LiDAR, radar, and camera collection across two vehicles. Initial qualitative inspection showed that LiDAR point clouds did not visibly change under CARLA weather settings. Radar was also sparse and initially difficult to inspect. The radar configuration was therefore revised to increase point density and to use four radar sensors: front-left, front-right, back-left, and back-right. These radars were oriented diagonally to cover a larger region around the ego vehicle and to provide front/back and left/right consistency comparisons. Later experiments focused on one ego vehicle and repeated weather regimes across matched spawn locations, allowing weather condition changes to be compared while reducing scene variation.

C. Feature Construction

All models operate on engineered statistical representations of the raw sensor data. The utilization of engineered statistical representations, rather than high-dimensional raw data, facilitates a targeted analysis of underlying signal characteristics and their sensitivity to environmental variance.

1) Camera Features: Camera features summarize grayscale intensity statistics: mean, standard deviation, min, max, quartiles, Laplacian variance as a blur proxy, dark-pixel ratio, and bright-pixel ratio. These features represent visibility and contrast changes expected under fog and rain. Camera features serve as a positive control because CARLA visually renders weather effects in RGB images.

2) LiDAR Features: LiDAR features summarize point-cloud geometry. For each frame, radial distance from the sensor is computed for all points. The feature vector includes point count, mean range, range standard deviation, minimum and maximum range, range percentiles, far-return ratio, and near-return ratio. These features are motivated by prior studies on atmospheric effects in active sensing. In the presence of fog

and rain, optical and near-infrared signals undergo attenuation and scattering, which reduces effective sensing range and alters return distributions. This behavior is commonly modeled using exponential attenuation (Beer–Lambert law[11]), and has been empirically observed in LiDAR degradation studies under adverse weather conditions [5], [9].

3) CARLA Radar Features: CARLA radar features were developed in stages. The basic feature set used point count, depth mean, depth standard deviation, minimum and maximum depth, azimuth sector counts, and depth histograms. The extended feature set added far-return ratio, near-return ratio, angular spread, angular irregularity, normalized density, depth standard deviation, and depth variance. After the radar configuration was expanded to four radars, multi-radar features were added: front/back mean-depth disagreement and left/right mean-depth disagreement. A temporal feature was also added by measuring frame-to-frame change in mean radar depth. These features are intended to approximate physical effects such as attenuation, scattering, and distributional shifts in sensor returns. In radar systems, precipitation and environmental clutter introduce variations in backscatter intensity and spatial distribution, while LiDAR measurements are affected by absorption and multi-path scattering phenomena. These effects have been documented in prior work on adverse-weather perception and sensor modeling [9], [10].

4) RADIATE Radar Features: RADIATE radar data was processed from Navtech polar radar images. Unlike CARLA radar detections, RADIATE provides real radar intensity images. Basic RADIATE features included global mean, standard deviation, min, max, quartiles, strong-reflection ratio, low-intensity ratio, and total energy. The extended feature set added range-bin statistics by splitting polar images into row-wise radial bins and angular variation statistics by measuring column-wise standard deviation. This maps the CARLA feature intent to the polar radar representation: rows correspond to range structure and columns correspond to angular structure.

D. Model

All classification experiments use a multilayer perceptron (MLP) with two hidden layers and ReLU activations. The output layer predicts one of three weather classes: clear, rain, or fog. Cross-entropy loss is used for classification. Feature normalization is computed using training-set statistics only and then applied to validation data. This prevents information leakage from validation distributions into the model.

E. Evaluation Protocols

CARLA experiments use run-level splits where possible: Entire recording sequences were partitioned into disjoint training and validation sets to mitigate the risk of temporal data leakage between highly correlated consecutive frames. RADIATE experiments initially used held-out frames when only one sequence per class was available. The final evaluation utilizes a train/validation split across independent sequences to ensure generalized performance metrics.

TABLE I
CARLA WEATHER CLASSIFICATION RESULTS

Setup	Feature Source	Best Val. Acc.
LiDAR-only	Distance statistics	0.33
Radar-only	Extended radar statistics	0.34
Camera-only	Intensity/visibility statistics	0.99
Fusion	Camera + radar	0.87–0.92

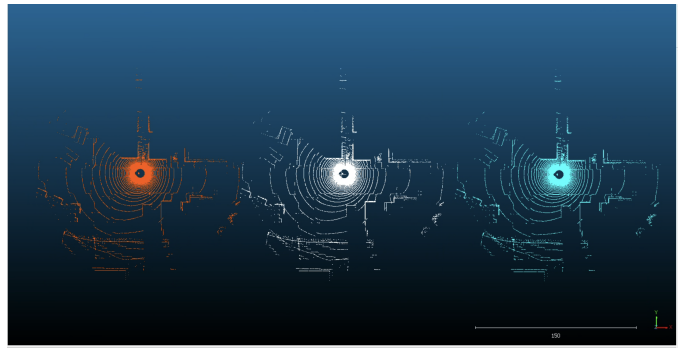

Fig. 1. Qualitative comparison of CARLA LiDAR point clouds under fog (orange), clear (white), and rain (blue). The point-cloud structure remains nearly unchanged across weather conditions, indicating a lack of simulated LiDAR degradation.

IV. Evaluation

A. Leakage and Sanity Checks

Before interpreting results, multiple checks were performed. First, training labels were randomly permuted. This permutation test resulted in validation accuracy converging to the theoretical chance baseline (0.33), confirming that the classification pipeline did not benefit from inadvertent label leakage. Second, train and validation runs were printed and verified to be disjoint for CARLA experiments. Third, train accuracy and validation accuracy were logged together to detect overfitting. Finally, RADIATE was restructured into explicit train and validation folders to avoid evaluating on adjacent frames from the same sequence.

B. CARLA Sensor Results

Table I summarizes the main CARLA experiments. Camera features achieved high validation accuracy, confirming that CARLA weather is strongly visible in RGB images. Radar-only and LiDAR-only results remained near chance. Fusion did not provide evidence that radar or LiDAR improved weather detection; performance was dominated by camera-derived signal.

The LiDAR result is supported qualitatively by point-cloud inspection and cloud-to-cloud comparison. In the tested CARLA configuration, point clouds under clear, fog, and rain remain visually similar, indicating that rendered weather does not produce observable LiDAR degradation.

C. CARLA Radar Feature Ablation

Radar feature ablations were performed to determine whether any feature family contained separable signal. The

TABLE II
CARLA RADAR FEATURE ABLATION. STOCHASTIC PERFORMANCE BASELINES ACROSS FEATURE GROUPS SUGGESTS THAT THE RADAR DATA LACKS WEATHER-DISCRIMINATIVE VARIATION IN THE TESTED SIMULATION CONFIGURATION.

Feature Group	Input Dim.	Best Val. Acc.
Density + depth stats	20	0.318
Basic + depth histogram + near/far ratios	48	0.327
Angular-only	8	0.334
All extended radar features	97 (fusion), 87 (radar-only)	0.34

TABLE III
RADIATE RADAR CLASSIFICATION RESULTS. SEPARATE TRAIN AND VALIDATION SEQUENCES WERE USED IN THE FINAL EXPERIMENT.

Feature Set	Train Acc. Trend	Best Val. Acc.
Basic intensity statistics	–	0.426
Extended range/angular features	up to $\sim$0.96	0.66

minimal feature set included density and depth statistics. Distribution features added depth histograms and near/far return ratios. Angular-only features tested azimuth spread and angular spacing. Extended features combined static, distributional, angular, multi-radar, and temporal summaries.

Ablation results across density, distributional, and angular feature groups yielding accuracies between 0.318 and 0.34 confirm a systemic lack of weather-discriminative variance in default CARLA radar outputs. Even with an increased input dimension of 97, performance remained indistinguishable from a random baseline.

D. RADIATE Radar Results

To test whether radar itself is incapable of weather inference or whether the limitation is simulation-specific, the same experimental logic was applied to RADIATE radar data. RADIATE provides real-world radar collected under adverse conditions and is therefore useful for checking whether weather-dependent radar signal exists outside simulation.

The first RADIATE experiment used basic image statistics and achieved approximately 0.426 validation accuracy. The extended feature set, which included range-bin and angular variation features, achieved validation accuracies around 0.66 across repeated runs. Table III summarizes these results.

The train-validation gap in the extended RADIATE experiment indicates overfitting, which is expected given the limited number of sequences and the compact MLP model. However, validation performance remains substantially above chance and above the CARLA radar baseline. This indicates that real radar data contains measurable weather-dependent structure that was not present in the tested CARLA radar outputs.

V. DISCUSSION

The experiments indicate that weather-aware perception from simulation requires careful attention to sensor fidelity. A simulator may render convincing visual weather while leaving non-visual sensor streams largely unchanged. In this project, camera features from CARLA were highly predictive of weather, but CARLA radar and LiDAR features did not provide meaningful signal. Engineered feature representations, including range histograms, angular spread, multi-radar consistency, and temporal variation, did not resolve the CARLA radar limitation. In contrast, RADIATE radar features achieved substantially higher validation accuracy, especially when range and angular structure were included.

This does not imply that radar alone solves weather detection. Rather, it shows that real radar measurements contain weather-dependent structure that simple statistical features can partially exploit. It also shows that default CARLA radar should not be assumed to model weather degradation sufficiently for training or validating weather-aware perception systems. For simulation-based research, explicit weather corruption models or physics-based sensor models may be necessary before drawing conclusions about radar or LiDAR reliability.

VI. LIMITATIONS

The study uses compact statistical features rather than full neural radar or point-cloud representations. This is appropriate for isolating signal presence but may understate the performance of richer models. RADIATE results are based on a limited number of sequences and should be interpreted as evidence of separability, not as a final weather classification benchmark. Finally, the study focuses on weather type classification; severity regression requires reliable continuous labels or physically grounded proxy variables and is left for future work.

VII. CONCLUSION

This project evaluated whether weather-dependent signal is present in simulated and real sensor data. CARLA camera features strongly encode weather, but CARLA radar and LiDAR features remain near chance in the tested configuration. RADIATE radar features, especially extended range and angular features, achieve substantially higher accuracy, indicating that real radar contains weather-dependent structure absent from the tested CARLA radar outputs. The evidence highlights that weather-aware simulation environments must integrate physically grounded degradation models to ensure that synthetic training data accurately reflects the reliability challenges encountered in adverse real-world conditions.

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